Continuous Functions are Dense in L^1

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Introduction

The goal of this paper is to prove that continuous functions are dense in $L^1(\mathbb{R}^n)$, since it is difficult to do so on a first pass of measure theory. The statement of the result below is originally from Folland: [Fol99].

Theorem 1. *Let $f \in L^1(\mathbb{R}^n)$ and $\varepsilon > 0$ be given. Then there exists a simple function $\phi = \sum_{k=1}^{N} a_j \chi_{R_k}$ where each R_k is a product of intervals, such that*

$$\int_{\mathbb{R}^n} |f - \phi| \, dm < \varepsilon,$$

and there is a continuous function g that vanishes outside a bounded set such that

$$\int_{\mathbb{R}^n} |f - g| \, dm < \varepsilon.$$

Proof. For the first claim, since $f \in L^1(\mathbb{R}^n)$, we may approximate f by a sequence of simple functions $\{\psi_n\}$ such that $0 \leq |\psi_1| \leq |\psi_2| \leq \cdots \leq |f|$ and $\psi_n \to f$ pointwise. Since $|\psi_n - f| \leq |\psi_n| + |f| \leq 2|f| \in L^1(\mathbb{R}^n)$, it follows from the Dominated Convergence Theorem that

$$\lim_{n\to\infty} \int_{\mathbb{R}^n} |\psi_n - f| \, dm = 0,$$

and in particular, given $\varepsilon > 0$ there exists a simple function $\psi = \sum_{k=1}^{N} a_k \chi_{E_k}$ such that

$$\int_{\mathbb{R}^n} |f - \psi| \, dm < \frac{\varepsilon}{2}. \tag{1}$$

We may assume without loss of generality that the E_k are disjoint, and one can infer

$$m(E_k) = \frac{1}{|a_k|}\int_{\mathbb{R}^n} |\psi|\chi_{E_k}\, dm \leq \frac{1}{|a_k|}\int_{E_k} |\psi|\, dm \leq \frac{1}{|a_k|}\int_{\mathbb{R}^n} |f|\, dm < \infty.$$

Now for the above $\varepsilon > 0$, for each E_k, there exists a finite collection $\{R_{j,k}\}_{j=1}^{N_k}$ of disjoint rectangles whose sides are intervals such that [1]

$$m\left(E_k \triangle \bigcup_{j=1}^{N_k} R_{j,k}\right) < \frac{\varepsilon}{2N|a_k|}.$$

Now define $R_k = \bigcup_{j=1}^{N_k} R_{j,k}$, and set $\phi = \sum_{k=1}^{N} a_k \chi_{R_k}$. By the Triangle Inequality,

$$\begin{aligned}
\int_{\mathbb{R}^n} |\psi - \phi|\, dm &= \int_{\mathbb{R}^n} \left|\sum_{k=1}^{N} a_k\chi_{E_k} - \sum_{k=1}^{N} a_k\chi_{R_k}\right| dm = \int_{\mathbb{R}^n} \left|\sum_{k=1}^{N} a_k(\chi_{E_k} - \chi_{R_k})\right| dm \\
&\leq \sum_{k=1}^{N} |a_k| \int_{\mathbb{R}^n} |\chi_{E_k} - \chi_{R_k}|\, dm \\
&= \sum_{k=1}^{N} |a_k| m\left(E_k \triangle \bigcup_{j=1}^{N_k} R_{j,k}\right) < \sum_{k=1}^{N} \frac{\varepsilon}{2N} = \frac{\varepsilon}{2}.
\end{aligned}$$

Combining this with Equation 1, one obtains

$$\begin{aligned}
\int_{\mathbb{R}^n} |f - \phi|\, dm &\leq \int_{\mathbb{R}^n} |f - \psi|\, dm + \int_{\mathbb{R}^n} |\psi - \phi|\, dm \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,
\end{aligned}$$

which proves the first claim.

To prove the second, let $\varepsilon > 0$ be given. Then by the result we just proved, there exists a simple function $\phi = \sum_{j=1}^{n} a_j \chi_{R_j}$ where each R_j is a product of intervals, such that

$$\int_{\mathbb{R}^n} |f - \phi|\, dm < \frac{\varepsilon}{2}. \tag{2}$$

We now wish to approximate ϕ by a continuous function g. To do so, consider the indicator χ_{R_j}. Let $\varepsilon > 0$ be given. The regularity of the Lebesgue measure

[1] See Theorem 2.40(c) in Folland.

implies that, for each R_j, there exists an open set U_j (which is bounded since the R_j are) with $R_j \subset U_j$ and $m(U_j \setminus R_j) < \varepsilon/2$, and also a compact set K_j such that $K_j \subset R_j$ and $m(R_j \setminus K_j) < \varepsilon/2$. We infer, then, that

$$\begin{aligned} m(U_j \setminus K_j) &= m(U_j) - m(K_j) = (m(U_j) - m(R_j)) + (m(R_j) - m(K_j)) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

In particular, given $\varepsilon > 0$ we can, for each R_j, choose open U_j and compact K_j such that $m(U_j \setminus K_j) < \frac{\varepsilon}{2|a_j|n}$. For each R_j, define

$$g_j(x) = 1 - \frac{d(x, K_j)}{d(x, K_j) + d(x, U_j^c)}.$$

Then g_j is continuous, $g(x) = 1$ when $x \in K_j$, and $0 \leq g(x) \leq 1$ on $U_j \setminus K_j$, and $g = 0$ on U_j^c. Write

$$\begin{aligned} \int_{\mathbb{R}^n} |\chi_{R_j} - g_j| \, dm &= \int_{K_j} |\chi_{R_j} - g_j| \, dm + \int_{U_j \setminus K_j} |\chi_{R_j} - g_j| \, dm + \int_{U_j^c} |\chi_{R_j} - g_j| \, dm \\ &= \int_{U_j \setminus K_j} |\chi_{R_j} - g_j| \, dm \leq \int_{U_j \setminus K_j} dm = m(U_j \setminus K_j) < \frac{\varepsilon}{2n|a_j|}. \end{aligned}$$

Defining $g(x) = \sum_{j=1}^n a_j g_j(x)$, we see that g is continuous, vanishes outside a bounded set, and

$$\int_{\mathbb{R}^n} |\phi - g| \, dm \leq \sum_{j=1}^n |a_j| \int_{\mathbb{R}^n} |\chi_{R_j} - g_j| \, dm < \sum_{j=1}^n |a_j| \left(\frac{\varepsilon}{2n|a_j|} \right) = \frac{\varepsilon}{2}. \tag{3}$$

Combining inequalities (2) and (3), we obtain the estimate

$$\int_{\mathbb{R}^n} |f - g| \, dm \leq \int_{\mathbb{R}^n} |f - \phi| \, dm + \int_{\mathbb{R}^n} |\phi - g| \, dm < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

which is precisely the statement of the Theorem. □

References

[Fol99] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. 2nd ed. Vol. 234. Pure and Applied Mathematics. New York: John Wiley & Sons, 1999. ISBN: 978-0-471-31716-6.

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